Lesson 7: Inner Products and Spectral Theory

AI/ML Mathematics

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1 Motivation and Overview

Suppose we wish to approximate a value function V^π in reinforcement learning using a linear combination of d basis functions $\phi_1, \ldots, \phi_d$. Among all vectors in the subspace $U = \text{span}(\phi_1, \ldots, \phi_d)$, which one is closest to V^π? This question has no answer without a notion of "distance," and distance requires an inner product. Once we equip the space with an inner product, the best approximation is the *orthogonal projection* of V^π onto U – a result that underpins least-squares methods, principal component analysis, and the natural policy gradient.

Beyond approximation, inner products unlock the most powerful structural theorem in finite-dimensional linear algebra: the *spectral theorem* for real symmetric matrices. This theorem guarantees that every real symmetric matrix has an orthonormal basis of eigenvectors with real eigenvalues – a fact that is foundational for PCA, kernel methods, Hessian analysis, and the Fisher information matrix in natural gradient methods. The tools developed in this lesson will let us define inner products, prove the spectral theorem, characterize positive definiteness, and establish the Rayleigh quotient bounds that govern convergence rates.

Where Inner Products and Spectral Theory Appear in RL/ML

- **Least squares and projection.** Linear value function approximation finds $\hat{V} = \Phi w$ minimizing $\|V^\pi - \Phi w\|_2^2$; the solution is the orthogonal projection of V^π onto $\text{range}(\Phi)$.
- **PCA and spectral decomposition.** Principal component analysis diagonalizes the covariance matrix $\Sigma = \frac{1}{n}\sum_i x_i x_i^\top$ using the spectral theorem to find the directions of maximum variance.
- **Condition number and convergence.** The condition number $\kappa(A) = \|A\| \cdot \|A^{-1}\|$ governs the convergence rate of iterative methods; for symmetric positive definite A, $\kappa = \lambda_{\max}/\lambda_{\min}$.
- **Natural policy gradient.** The Fisher information matrix $F(\theta)$ defines an inner product on the parameter space; the natural gradient $\tilde{\nabla} J = F^{-1}\nabla J$ is the steepest ascent direction in this geometry, computed via spectral decomposition.

Prerequisites. Determinants, eigenvalues, characteristic polynomial, algebraic and geometric multiplicity, diagonalizability criterion, Cayley–Hamilton theorem (Lesson 6).

Learning objectives. After completing this lesson, the student will be able to:

- **Define** inner products, norms, and orthogonality, and verify the inner product axioms in concrete examples.
- **Prove** the Cauchy–Schwarz inequality and apply the Gram–Schmidt process to construct orthonormal bases.
- **Derive** the orthogonal projection formula and prove the best approximation theorem.
- **Prove** the spectral theorem for real symmetric matrices: every such matrix is orthogonally diagonalizable.
- **Classify** symmetric matrices as positive definite, positive semidefinite, or indefinite using eigenvalue criteria.
- **Apply** the Rayleigh quotient and Courant–Fischer theorem to characterize extreme eigenvalues.

2 References

- Axler [2015, Chapters 6–7] – Inner product spaces, orthonormal bases, the spectral theorem for self-adjoint operators.

- Hoffman and Kunze [1971, Chapter 8] – Inner product spaces and the principal axis theorem with detailed proofs.
- Strang [2016, Chapters 6–7] – Computational perspective on orthogonality, Gram–Schmidt, and symmetric matrices.
- Horn and Johnson [2013, Chapter 4] – Spectral theory for Hermitian matrices, Rayleigh quotient, Courant–Fischer, and Weyl inequalities.
- Lax [2007, Chapter 8] – Spectral theory with connections to functional analysis and operator theory.

3 Inner Product Spaces

3.1 Definition and Examples

Definition 3.1 (Inner Product)**.** An *inner product* on a real vector space V is a function $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{R}$ satisfying, for all $u, v, w \in V$ and $\alpha \in \mathbb{R}$:

1. **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$.
2. **Linearity in the first argument:** $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$.
3. **Positive definiteness:** $\langle v, v \rangle \geq 0$, with equality if and only if $v = 0$.

A vector space equipped with an inner product is called an *inner product space*.

Remark 3.2. Symmetry and linearity in the first argument together imply linearity in the second argument (bilinearity): $\langle u, \alpha v + w \rangle = \langle \alpha v + w, u \rangle = \alpha \langle v, u \rangle + \langle w, u \rangle = \alpha \langle u, v \rangle + \langle u, w \rangle$.

Example 3.3 (Standard Inner Products)**.**

1. **Euclidean inner product on $\mathbb{R}^n$:** $\langle x, y \rangle = \sum_{i=1}^n x_i y_i = x^\top y$.
2. **Weighted inner product on $\mathbb{R}^n$:** For a symmetric positive definite matrix $M \in \mathbb{R}^{n \times n}$, $\langle x, y \rangle_M = x^\top M y$ defines an inner product.
3. **L^2 inner product on $C[a, b]$:** $\langle f, g \rangle = \int_a^b f(t) g(t)\, dt$ for continuous functions $f, g \colon [a, b] \to \mathbb{R}$.

Definition 3.4 (Norm Induced by an Inner Product)**.** The *norm* induced by an inner product is $\|v\| = \sqrt{\langle v, v \rangle}$. The *distance* between u and v is $\|u - v\|$.

3.2 Cauchy–Schwarz and Triangle Inequalities

Theorem 3.5 (Cauchy–Schwarz Inequality)**.** *For all u, v in an inner product space,*

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|.$$

Equality holds if and only if u and v are linearly dependent.

Proof. If $v = 0$, both sides are zero. Assume $v \neq 0$. For any $t \in \mathbb{R}$,

$$0 \leq \|u - tv\|^2 = \langle u - tv, u - tv \rangle = \|u\|^2 - 2t\langle u, v \rangle + t^2 \|v\|^2.$$

This is a quadratic in t that is nonneg for all t. Minimizing over t at $t^* = \langle u, v \rangle / \|v\|^2$ gives

$$0 \leq \|u\|^2 - \frac{\langle u, v \rangle^2}{\|v\|^2},$$

which rearranges to $\langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2$. Taking square roots yields $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$.

Equality holds iff $\|u - t^* v\| = 0$, i.e., $u = t^* v$ (linear dependence). $\square$

Corollary 3.6 (Triangle Inequality). *For all u, v in an inner product space, $\|u+v\| \leq \|u\|+\|v\|$.*

Proof.

$$\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\,\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2. \qquad \square$$

With the Cauchy–Schwarz inequality established, we can develop the geometric structure of inner product spaces: orthogonality, projections, and orthonormal bases.

4 Orthogonality and Orthonormal Bases

4.1 Orthogonal and Orthonormal Lists

Definition 4.1 (Orthogonality). Two vectors u, v in an inner product space are *orthogonal*, written $u \perp v$, if $\langle u, v\rangle = 0$. A list of vectors $v_1, \ldots, v_m$ is *orthogonal* if $\langle v_i, v_j\rangle = 0$ for all $i \neq j$, and *orthonormal* if additionally $\|v_i\| = 1$ for each i.

Theorem 4.2 (Pythagorean Theorem). *If $v_1, \ldots, v_m$ are pairwise orthogonal, then*

$$\|v_1 + \cdots + v_m\|^2 = \|v_1\|^2 + \cdots + \|v_m\|^2.$$

Proof. $\|v_1+\cdots+v_m\|^2 = \langle \sum_i v_i, \sum_j v_j\rangle = \sum_i \sum_j \langle v_i, v_j\rangle = \sum_i \|v_i\|^2$, since all cross terms vanish by orthogonality. $\square$

Proposition 4.3 (Orthogonal Lists Are Independent). *Any orthogonal list of nonzero vectors is linearly independent.*

Proof. Let $v_1, \ldots, v_m$ be orthogonal and nonzero. If $c_1 v_1 + \cdots + c_m v_m = 0$, take the inner product with v_k: $c_k\|v_k\|^2 = 0$. Since $\|v_k\| \neq 0$, $c_k = 0$ for each k. $\square$

Example 4.4 (Orthonormal Basis of $\mathbb{R}^3$). The standard basis $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ is orthonormal with respect to the Euclidean inner product. For any $v = (a, b, c) \in \mathbb{R}^3$, the coordinates are recovered by inner products: $a = \langle v, e_1\rangle$, $b = \langle v, e_2\rangle$, $c = \langle v, e_3\rangle$.

4.2 The Gram–Schmidt Process

Theorem 4.5 (Gram–Schmidt Orthogonalization). *Let $v_1, \ldots, v_m$ be a linearly independent list in an inner product space V. Then there exists an orthonormal list $e_1, \ldots, e_m$ such that*

$$\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(v_1, \ldots, v_k) \quad \textit{for each } k = 1, \ldots, m.$$

The orthonormal list is constructed by:

$$\begin{aligned} &w_1 = v_1, \quad e_1 = w_1/\|w_1\|, \\ &w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j\rangle\, e_j, \quad e_k = w_k/\|w_k\|, \quad k = 2, \ldots, m. \end{aligned}$$

Proof. We proceed by induction. **Base case:** $v_1 \neq 0$ (by independence), so $w_1 = v_1 \neq 0$ and $e_1 = v_1/\|v_1\|$ is a unit vector. Clearly $\operatorname{span}(e_1) = \operatorname{span}(v_1)$.

Inductive step: Assume $e_1, \ldots, e_{k-1}$ is orthonormal with $\operatorname{span}(e_1, \ldots, e_j) = \operatorname{span}(v_1, \ldots, v_j)$ for $j \leq k-1$. Define $w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j\rangle\, e_j$. For each $i \leq k-1$:

$$\langle w_k, e_i\rangle = \langle v_k, e_i\rangle - \sum_{j=1}^{k-1} \langle v_k, e_j\rangle \langle e_j, e_i\rangle = \langle v_k, e_i\rangle - \langle v_k, e_i\rangle = 0.$$

Also $w_k \neq 0$: if $w_k = 0$, then $v_k = \sum_j \langle v_k, e_j \rangle e_j \in \operatorname{span}(e_1, \ldots, e_{k-1}) = \operatorname{span}(v_1, \ldots, v_{k-1})$, contradicting the independence of the original list. Set $e_k = w_k / \|w_k\|$. Then $e_1, \ldots, e_k$ is orthonormal, and $\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(v_1, \ldots, v_k)$ (since w_k is a combination of v_k and $e_1, \ldots, e_{k-1}$, and v_k can be recovered from w_k and $e_1, \ldots, e_{k-1}$). □

Example 4.6 (Gram–Schmidt in $\mathbb{R}^3$)**.** Apply Gram–Schmidt to $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$.

Step 1: $w_1 = (1, 1, 0)$, $\|w_1\| = \sqrt{2}$, $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

Step 2: $\langle v_2, e_1 \rangle = 1/\sqrt{2}$. $w_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = (1/2, -1/2, 1)$. $\|w_2\| = \sqrt{3/2}$, $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$. One verifies $\langle e_1, e_2 \rangle = 0$ and $\operatorname{span}(e_1, e_2) = \operatorname{span}(v_1, v_2)$.

Corollary 4.7. *Every finite-dimensional inner product space has an orthonormal basis.*

Proof. Start with any basis and apply the Gram–Schmidt process (Theorem 4.5). □

4.3 Orthogonal Complement and Decomposition

Definition 4.8 (Orthogonal Complement)**.** If U is a subspace of an inner product space V, the *orthogonal complement* of U is

$$U^\perp = \{v \in V : \langle v, u \rangle = 0 \text{ for all } u \in U\}.$$

Proposition 4.9. *$U^\perp$ is a subspace of V, and $U \cap U^\perp = \{0\}$.*

Proof. $U^\perp$ is a subspace: $0 \in U^\perp$, and closure under linear combinations follows from bilinearity. If $v \in U \cap U^\perp$, then $\langle v, v \rangle = 0$, so $v = 0$. □

Theorem 4.10 (Orthogonal Decomposition)**.** *If V is a finite-dimensional inner product space and U is a subspace, then $V = U \oplus U^\perp$. Equivalently, every $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in U^\perp$.*

Proof. Let $e_1, \ldots, e_k$ be an orthonormal basis of U (exists by Corollary 4.7). For any $v \in V$, set

$$u = \sum_{j=1}^{k} \langle v, e_j \rangle e_j, \qquad w = v - u.$$

Then $u \in U$ and for each $i = 1, \ldots, k$:

$$\langle w, e_i \rangle = \langle v, e_i \rangle - \sum_{j=1}^{k} \langle v, e_j \rangle \langle e_j, e_i \rangle = \langle v, e_i \rangle - \langle v, e_i \rangle = 0.$$

Hence $w \perp e_i$ for each basis element of U, so $w \in U^\perp$. This shows $V = U + U^\perp$. By Proposition 4.9, $U \cap U^\perp = \{0\}$, so the sum is direct. □

Corollary 4.11 (Dimension Formula)**.** $\dim U + \dim U^\perp = \dim V$ *and* $(U^\perp)^\perp = U$.

Proof. The first follows from $V = U \oplus U^\perp$. For the second, $U \subseteq (U^\perp)^\perp$ is clear, and both have dimension $\dim U$. □

4.4 Orthogonal Projection and Best Approximation

Definition 4.12 (Orthogonal Projection)**.** Let U be a subspace of a finite-dimensional inner product space V. The *orthogonal projection* onto U is the map $\operatorname{proj}_U \colon V \to V$ defined by $\operatorname{proj}_U(v) = u$, where $v = u + w$ is the orthogonal decomposition with $u \in U$ and $w \in U^\perp$. If $e_1, \ldots, e_k$ is an orthonormal basis of U, then

$$\operatorname{proj}_U(v) = \sum_{j=1}^{k} \langle v, e_j \rangle e_j.$$

Orthogonal Projection and Best Approximation

Theorem 4.13 (Best Approximation)**.** *Let U be a subspace of a finite-dimensional inner product space V and let $v \in V$. Then $\operatorname{proj}_U(v)$ is the unique element of U that minimizes the distance to v:*

$$\|v - \operatorname{proj}_U(v)\| < \|v - u\| \quad \text{for all } u \in U,\ u \neq \operatorname{proj}_U(v).$$

Proof. Let $\hat{u} = \operatorname{proj}_U(v)$, so $v - \hat{u} \in U^\perp$. For any $u \in U$:

$$\|v - u\|^2 = \|(v - \hat{u}) + (\hat{u} - u)\|^2 = \|v - \hat{u}\|^2 + \|\hat{u} - u\|^2,$$

where the cross term vanishes because $v - \hat{u} \in U^\perp$ and $\hat{u} - u \in U$ (Pythagorean theorem, Theorem 4.2). Hence $\|v - u\|^2 \geq \|v - \hat{u}\|^2$, with equality iff $u = \hat{u}$. □

Example 4.14 (Projection onto a Line)**.** Let $U = \operatorname{span}(a)$ for $a \neq 0$ in $\mathbb{R}^n$. The orthonormal basis of U is $e_1 = a/\|a\|$, so

$$\operatorname{proj}_U(v) = \frac{\langle v, a\rangle}{\|a\|^2}\, a = \frac{a^\top v}{a^\top a}\, a.$$

For $a = (1,1)^\top$ and $v = (3,1)^\top$ in $\mathbb{R}^2$: $\operatorname{proj}_U(v) = \frac{4}{2}(1,1)^\top = (2,2)^\top$. The residual $v - \operatorname{proj}_U(v) = (1,-1)^\top$ is orthogonal to a: $(1,-1)\cdot(1,1) = 0$.

Least Squares and Linear Value Function Approximation

In linear value function approximation, states are represented by feature vectors $\phi(s_i) \in \mathbb{R}^d$, forming a matrix $\Phi \in \mathbb{R}^{n\times d}$. The projected Bellman equation seeks $w^* \in \mathbb{R}^d$ such that Φw^* is the orthogonal projection of $r + \gamma PV$ onto $\operatorname{range}(\Phi)$ under the inner product $\langle u, v\rangle_\mu = \sum_i \mu(s_i) u_i v_i$ weighted by the stationary distribution μ. By the best approximation theorem (Theorem 4.13), this projection minimizes $\|V - \Phi w\|_\mu^2$ over all $w \in \mathbb{R}^d$. The normal equations $\Phi^\top D_\mu \Phi\, w = \Phi^\top D_\mu (r + \gamma PV)$ are the concrete form of $v - \operatorname{proj}_U(v) \in U^\perp$.

4.5 Bessel's Inequality

Theorem 4.15 (Bessel's Inequality)**.** *Let $e_1, \ldots, e_m$ be an orthonormal list (not necessarily a basis) in an inner product space V. For any $v \in V$,*

$$\sum_{j=1}^{m} |\langle v, e_j\rangle|^2 \leq \|v\|^2.$$

Equality holds if and only if $v \in \operatorname{span}(e_1, \ldots, e_m)$.

Proof. With $U = \operatorname{span}(e_1, \ldots, e_m)$, the Pythagorean theorem gives $\|v\|^2 = \|\operatorname{proj}_U(v)\|^2 + \|v - \operatorname{proj}_U(v)\|^2 = \sum_j |\langle v, e_j\rangle|^2 + \|v - \operatorname{proj}_U(v)\|^2 \geq \sum_j |\langle v, e_j\rangle|^2$, with equality iff $v \in U$. □

5 The Spectral Theorem for Real Symmetric Matrices

We now turn to the central result of this lesson. Throughout this section, all matrices are real ($\mathbb{F} = \mathbb{R}$) and the inner product is the standard Euclidean inner product $\langle x, y\rangle = x^\top y$.

5.1 Eigenvalues of Symmetric Matrices Are Real

Lemma 5.1. *Every eigenvalue of a real symmetric matrix $A \in \mathbb{R}^{n\times n}$ is real.*

Proof. We work over $\mathbb{C}$ temporarily. Let $\lambda \in \mathbb{C}$ be an eigenvalue with eigenvector $v \in \mathbb{C}^n \setminus \{0\}$: $Av = \lambda v$. Take the conjugate transpose: $\bar{v}^\top A^\top = \bar{\lambda}\bar{v}^\top$. Since $A = A^\top$ (and A has real entries, so $\bar{A} = A$):

$$\bar{v}^\top Av = \bar{\lambda}\,\bar{v}^\top v.$$

Also, $\bar{v}^\top Av = \bar{v}^\top(\lambda v) = \lambda\,\bar{v}^\top v$. Since $\bar{v}^\top v = \sum_i |v_i|^2 > 0$, we can divide to get $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$. □

5.2 Orthogonality of Eigenvectors

Lemma 5.2. *Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Proof. Let $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$. Then

$$\lambda_1\langle v_1, v_2\rangle = \langle Av_1, v_2\rangle = v_1^\top A^\top v_2 = v_1^\top Av_2 = \langle v_1, Av_2\rangle = \lambda_2\langle v_1, v_2\rangle.$$

Since $\lambda_1 \neq \lambda_2$, we must have $\langle v_1, v_2\rangle = 0$. □

5.3 Invariant Subspaces and the Spectral Theorem

Lemma 5.3 (Symmetric Matrices Preserve Orthogonal Complements)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric and let U be an A-invariant subspace (i.e., $Au \in U$ for all $u \in U$). Then $U^\perp$ is also A-invariant.*

Proof. Let $w \in U^\perp$. For any $u \in U$, we have $Au \in U$ (by A-invariance) and

$$\langle Aw, u\rangle = \langle w, A^\top u\rangle = \langle w, Au\rangle = 0,$$

since $Au \in U$ and $w \in U^\perp$. Hence $Aw \perp U$, i.e., $Aw \in U^\perp$. □

The Spectral Theorem for Real Symmetric Matrices

Theorem 5.4 (Spectral Theorem)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then:*

1. *All eigenvalues of A are real.*
2. *There exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A.*
3. *There exists an orthogonal matrix Q (i.e., $Q^\top Q = I$) such that $Q^\top AQ = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.*

Equivalently, A admits the spectral decomposition *$A = \sum_{i=1}^n \lambda_i\, q_i q_i^\top$, where $q_1, \ldots, q_n$ are orthonormal eigenvectors.*

Proof. (1) is Lemma 5.1.

(2) We prove by induction on n. For $n = 1$, any nonzero vector is an eigenvector, and normalizing gives an orthonormal basis.

For the inductive step, let $n \geq 2$. Since the characteristic polynomial p_A has degree n and real coefficients, and all roots are real (by part (1)), A has at least one real eigenvalue λ_1 with unit eigenvector q_1. Let $U = \operatorname{span}(q_1)$. Since $AU = \lambda_1 U \subseteq U$, the subspace U is A-invariant. By Lemma 5.3, $U^\perp$ is also A-invariant.

Let $B = A|_{U^\perp}$ denote the restriction of A to $U^\perp$. Since $U^\perp$ has dimension $n-1$ and B is symmetric (as the restriction of a symmetric operator to an invariant subspace), the inductive hypothesis provides an orthonormal basis $q_2, \ldots, q_n$ of $U^\perp$ consisting of eigenvectors of B (and

hence of A). Then $q_1, q_2, \ldots, q_n$ is an orthonormal basis of $\mathbb{R}^n = U \oplus U^\perp$ consisting of eigenvectors of A.

(3) Setting $Q = (q_1 \mid q_2 \mid \cdots \mid q_n)$, the columns are orthonormal, so $Q^\top Q = I$. The eigenvector equation $AQ = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ gives $Q^\top AQ = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

The spectral decomposition follows from $A = Q\Lambda Q^\top$ where $\Lambda = \mathrm{diag}(\lambda_i)$: $A = \sum_{i=1}^n \lambda_i\, q_i q_i^\top$. □

Example 5.5 (Spectral Decomposition of a 2×2 Matrix)**.** Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. The characteristic polynomial is $\lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4)$. Eigenvalues: $\lambda_1 = 2$, $\lambda_2 = 4$.

For $\lambda_1 = 2$: $q_1 = \frac{1}{\sqrt{2}}(1, -1)^\top$. For $\lambda_2 = 4$: $q_2 = \frac{1}{\sqrt{2}}(1, 1)^\top$.

The spectral decomposition is $A = 2q_1q_1^\top + 4q_2q_2^\top = 2 \cdot \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + 4 \cdot \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

Example 5.6 (Spectral Decomposition of a 3×3 Matrix)**.** Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Expanding $\det(A - \lambda I) = (2 - \lambda)[(2 - \lambda)^2 - 2]$ gives eigenvalues $\lambda_1 = 2 - \sqrt{2}$, $\lambda_2 = 2$, $\lambda_3 = 2 + \sqrt{2}$ – all real and distinct, confirming the spectral theorem.

PCA and Spectral Decomposition

In principal component analysis, given centered data $x_1, \ldots, x_m \in \mathbb{R}^n$, the sample covariance matrix is $S = \frac{1}{m}\sum_{i=1}^m x_i x_i^\top$. Since S is symmetric and positive semidefinite, the spectral theorem gives $S = \sum_{j=1}^n \lambda_j q_j q_j^\top$ with $\lambda_j \geq 0$. The top k principal components are the eigenvectors $q_1, \ldots, q_k$ (ordered by $\lambda_1 \geq \cdots \geq \lambda_n$). The projection $\mathrm{proj}_U(x) = \sum_{j=1}^k \langle x, q_j \rangle q_j$ onto $U = \mathrm{span}(q_1, \ldots, q_k)$ is the best rank-k approximation of the data by the Eckart–Young theorem.

6 Positive Definiteness and Cholesky Factorization

6.1 Characterizations of Positive Definiteness

Definition 6.1 (Positive Definite and Positive Semidefinite)**.** A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is:

- *Positive definite* (written $A \succ 0$) if $x^\top Ax > 0$ for all nonzero $x \in \mathbb{R}^n$.
- *Positive semidefinite* (written $A \succeq 0$) if $x^\top Ax \geq 0$ for all $x \in \mathbb{R}^n$.

Theorem 6.2 (Characterizations of Positive Definiteness)**.** *For a symmetric matrix $A \in \mathbb{R}^{n \times n}$, the following are equivalent:*

1. *$A \succ 0$ (positive definite).*
2. *All eigenvalues of A are strictly positive.*
3. *$A = R^\top R$ for some invertible matrix R.*
4. *All leading principal minors of A are strictly positive.*

Proof. (1) $\Rightarrow$ (2): If $Av = \lambda v$ with $v \neq 0$, then $0 < v^\top Av = \lambda \|v\|^2$, so $\lambda > 0$.

(2) $\Rightarrow$ (3): By the spectral theorem, $A = Q\Lambda Q^\top$ with $\lambda_i > 0$. Set $R = \Lambda^{1/2}Q^\top$; then $R^\top R = A$ and R is invertible.

(3) $\Rightarrow$ (1): $x^\top Ax = \|Rx\|^2 > 0$ for $x \neq 0$ since R is invertible.

(1) $\Leftrightarrow$ (4): See Horn and Johnson [2013, Theorem 7.2.5]; the key idea is that principal submatrices of positive definite matrices are positive definite. □

Positive Definiteness Requires Symmetry

The non-symmetric matrix $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ has eigenvalues $\{1, 1\}$ (positive), yet $x^\top Ax = (x_1 + x_2)^2 = 0$ at $x = (1, -1)^\top$. The eigenvalue characterization in Theorem 6.2 requires symmetry.

6.2 Cholesky Factorization

Theorem 6.3 (Cholesky Factorization)**.** *If $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, then there exists a unique lower triangular matrix L with positive diagonal entries such that $A = LL^\top$.*

Proof. **Existence** (by induction on n). For $n = 1$, $L = (\sqrt{a_{11}})$. For $n \geq 2$, partition A as $A = \left(\begin{smallmatrix} a_{11} & b^\top \\ b & C \end{smallmatrix}\right)$ with $a_{11} > 0$. Set $\ell_{11} = \sqrt{a_{11}}$ and $\ell = b/\ell_{11}$. The Schur complement $S = C - \ell\ell^\top$ is symmetric positive definite, so by induction $S = L_1 L_1^\top$. Then $L = \left(\begin{smallmatrix} \ell_{11} & 0 \\ \ell & L_1 \end{smallmatrix}\right)$ satisfies $A = LL^\top$.

Uniqueness: If $LL^\top = MM^\top$, then $Q = M^{-1}L$ satisfies $QQ^\top = I$. Since Q is lower triangular with positive diagonal (as a product of lower triangular matrices with positive diagonals), and orthogonal lower triangular matrices with positive diagonal must equal I, we get $L = M$. □

Example 6.4 (Cholesky Factorization)**.** Let $A = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$. Since $\det A = 16 > 0$ and $a_{11} = 4 > 0$, A is positive definite. $\ell_{11} = 2$, $\ell_{21} = 2/2 = 1$, $s_{22} = 5 - 1 = 4$, $\ell_{22} = 2$. So $L = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$ and $LL^\top = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix} = A$.

Condition Number and Convergence of Iterative Methods

For a symmetric positive definite system $Ax = b$, the condition number $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ determines the rate of convergence of gradient descent and conjugate gradient methods. Preconditioning replaces $Ax = b$ with $L^{-1}AL^{-\top}(L^\top x) = L^{-1}b$ using the Cholesky factor $A \approx LL^\top$ of a preconditioner. A well-chosen preconditioner reduces κ and accelerates convergence. In RL, the Fisher information matrix $F(\theta)$ is symmetric positive (semi)definite; its condition number affects the convergence of natural policy gradient methods.

7 Rayleigh Quotient and Courant–Fischer

7.1 The Rayleigh Quotient

Definition 7.1 (Rayleigh Quotient)**.** For a symmetric matrix $A \in \mathbb{R}^{n\times n}$, the *Rayleigh quotient* is the function $R_A \colon \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ defined by

$$R_A(x) = \frac{x^\top Ax}{x^\top x} = \frac{\langle Ax, x\rangle}{\|x\|^2}.$$

Theorem 7.2 (Extreme Values of the Rayleigh Quotient)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$. Then*

$$\lambda_1 = \min_{x\neq 0} R_A(x), \qquad \lambda_n = \max_{x\neq 0} R_A(x).$$

The minimum is attained at any eigenvector for λ_1, and the maximum at any eigenvector for λ_n.

Proof. Substituting $y = Q^\top x$ in the spectral decomposition $A = Q\Lambda Q^\top$: $R_A(x) = \sum_i \lambda_i y_i^2 / \sum_i y_i^2$, a weighted average of eigenvalues with nonneg weights summing to 1. Hence $\lambda_1 \le R_A(x) \le \lambda_n$. Equality is attained at the corresponding eigenvectors: $R_A(q_1) = \lambda_1$ and $R_A(q_n) = \lambda_n$. □

7.2 The Courant–Fischer Minimax Theorem

Theorem 7.3 (Courant–Fischer Minimax Theorem). *Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \le \cdots \le \lambda_n$. For each $k = 1, \ldots, n$,*

$$\lambda_k = \min_{\substack{S\subseteq\mathbb{R}^n \\ \dim S = k}} \max_{\substack{x\in S \\ x\neq 0}} R_A(x) = \max_{\substack{S\subseteq\mathbb{R}^n \\ \dim S = n-k+1}} \min_{\substack{x\in S \\ x\neq 0}} R_A(x).$$

Proof. We prove $\lambda_k = \min_{\dim S=k} \max_{x\in S\setminus\{0\}} R_A(x)$. Let $q_1, \ldots, q_n$ be orthonormal eigenvectors with $Aq_i = \lambda_i q_i$.

Upper bound: $S_0 = \operatorname{span}(q_1, \ldots, q_k)$ has dimension k. For $x = \sum_{i=1}^k c_i q_i \neq 0$, $R_A(x) = \sum_i \lambda_i c_i^2 / \sum_i c_i^2 \le \lambda_k$, with equality at $x = q_k$.

Lower bound: For any k-dimensional S, let $T = \operatorname{span}(q_k, \ldots, q_n)$ with $\dim T = n - k + 1$. Since $\dim S + \dim T > n$, there exists $x \in S \cap T \setminus \{0\}$. Writing $x = \sum_{i=k}^n c_i q_i$ gives $R_A(x) \ge \lambda_k$.

Combining: $\lambda_k = \min_{\dim S=k} \max_{x\in S} R_A(x)$. The max-min formula is proved analogously. □

Example 7.4 (Rayleigh Quotient Bounds). For $A = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$, evaluating R_A at $x = (0,1)^\top$ gives $R_A = 2$, so $\lambda_{\min} \le 2$. At $x = (1,1)^\top$, $R_A = 4$, so $\lambda_{\max} \ge 4$. The exact eigenvalues $3 \pm \sqrt{2} \approx 1.59, 4.41$ are consistent.

Natural Policy Gradient and the Fisher Matrix

In natural policy gradient methods, the update $\theta_{t+1} = \theta_t + \alpha F(\theta_t)^{-1}\nabla J(\theta_t)$ uses the Fisher information matrix $F(\theta) \succeq 0$. Its spectral decomposition enables efficient computation of $F^{-1}\nabla J$, and the Courant–Fischer theorem characterizes how the curvature of the KL-divergence landscape varies across parameter subspaces.

8 Exercises

Exercise 8.1 (Inner Product Verification). (∗)
Let $V = \mathbb{R}^2$ and define $\langle x, y\rangle = 2x_1y_1 + x_1y_2 + x_2y_1 + 3x_2y_2$.

(a) Verify that this is an inner product by checking all axioms.
(b) Find the matrix M such that $\langle x, y\rangle = x^\top My$.
(c) Compute $\|e_1\|$ and $\|e_2\|$ in this inner product.

Exercise 8.2 (Gram–Schmidt and Projection). (∗)
In $\mathbb{R}^3$ with the standard inner product:

(a) Apply the Gram–Schmidt process to $v_1 = (1,1,1)$, $v_2 = (1,1,0)$, $v_3 = (1,0,0)$ to produce an orthonormal basis.
(b) Compute the orthogonal projection of $w = (2,3,5)$ onto $U = \operatorname{span}(v_1, v_2)$.
(c) Verify that $w - \operatorname{proj}_U(w)$ is orthogonal to both v_1 and v_2.

Exercise 8.3 (Spectral Decomposition). (∗∗)
Let $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$.

(a) Find all eigenvalues and a corresponding orthonormal eigenbasis.
(b) Write the spectral decomposition $A = \lambda_1 q_1 q_1^\top + \lambda_2 q_2 q_2^\top$.

(c) Compute A^{10} using the spectral decomposition.
(d) Determine whether A is positive definite.

Exercise 8.4 (PCA Application). (∗∗)
Consider centered data points $x_1 = (2, 1)^\top$, $x_2 = (-1, 1)^\top$, $x_3 = (-1, -2)^\top$ in $\mathbb{R}^2$.
(a) Compute the sample covariance matrix $S = \frac{1}{3}\sum_{i=1}^{3} x_i x_i^\top$.
(b) Find the eigenvalues and eigenvectors of S.
(c) Identify the first principal component direction and compute the projection of each data point onto it.
(d) What fraction of the total variance is captured by the first principal component?

Exercise 8.5 (Courant–Fischer Application). (∗∗)
Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \le \cdots \le \lambda_n$.
(a) Use the Courant–Fischer theorem to prove the *interlacing inequality*: if B is the $(n-1)\times(n-1)$ principal submatrix obtained by deleting the last row and column of A, and B has eigenvalues $\mu_1 \le \cdots \le \mu_{n-1}$, then $\lambda_1 \le \mu_1 \le \lambda_2 \le \mu_2 \le \cdots \le \mu_{n-1} \le \lambda_n$.
(b) Apply this result: if $A \in \mathbb{R}^{3\times 3}$ is symmetric with eigenvalues $1, 3, 5$, what are the possible eigenvalues of the 2×2 upper-left submatrix?
Hint: For (a), embed $(n-1)$-dimensional subspaces of $\mathbb{R}^{n-1}$ into $\mathbb{R}^n$ via $x \mapsto (x, 0)$.

Exercise 8.6 (Positive Definiteness and Convexity). (∗ ∗ ∗)

(a) Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric with $A \succeq B$. Use the Courant–Fischer theorem to prove $\lambda_k(A) \ge \lambda_k(B)$ for each k.
(b) Let $A_0, A_1 \succ 0$ and $A(t) = (1-t)A_0 + tA_1$ for $t \in [0, 1]$. Show that $A(t) \succ 0$ for all t (the positive definite cone is convex).

9 Summary

- An *inner product* on a real vector space is a symmetric, bilinear, positive definite function $\langle\cdot,\cdot\rangle$. It induces a norm $\|v\| = \sqrt{\langle v, v\rangle}$ and a distance $d(u, v) = \|u - v\|$.
- The *Cauchy–Schwarz inequality* $|\langle u, v\rangle| \le \|u\| \cdot \|v\|$ is the most fundamental inequality in inner product spaces, with equality iff u and v are linearly dependent.
- The *Gram–Schmidt process* converts any linearly independent list into an orthonormal list with the same successive spans. Every finite-dimensional inner product space has an orthonormal basis.
- For any subspace U, the space decomposes as $V = U \oplus U^\perp$. The *orthogonal projection* $\mathrm{proj}_U(v) = \sum_j \langle v, e_j\rangle e_j$ gives the unique closest point in U to v (best approximation theorem).
- *Bessel's inequality*: $\sum_j |\langle v, e_j\rangle|^2 \le \|v\|^2$ for any orthonormal list, with equality iff v lies in the span.
- The *spectral theorem* states that every real symmetric matrix has an orthonormal basis of eigenvectors with real eigenvalues: $A = Q\Lambda Q^\top$ with Q orthogonal.
- A symmetric matrix is *positive definite* iff all eigenvalues are positive, iff $x^\top Ax > 0$ for all $x \ne 0$, iff it admits a Cholesky factorization $A = LL^\top$.
- The *Rayleigh quotient* $R_A(x) = x^\top Ax/x^\top x$ satisfies $\lambda_{\min} \le R_A(x) \le \lambda_{\max}$. The *Courant–Fischer theorem* gives a minimax characterization of every eigenvalue, enabling eigenvalue comparison between matrices and submatrices.

10 Further Reading

- **Lesson 8** (SVD and Applications) extends the spectral theorem to non-symmetric and non-square matrices via the singular value decomposition $A = U\Sigma V^\top$. The SVD is the most widely used matrix factorization in machine learning.
- **Phase 01, Functional Analysis** (Lessons on Hilbert spaces) generalizes the finite-dimensional inner product theory to infinite-dimensional spaces, with applications to kernel methods and reproducing kernel Hilbert spaces (RKHS).
- The complex analogue of the spectral theorem (for Hermitian matrices) is developed in Horn and Johnson [2013, Chapter 4], along with Weyl's eigenvalue perturbation inequalities.
- Axler [2015, Chapter 6] presents inner product spaces and the spectral theorem in the coordinate-free operator-theoretic framework, avoiding matrices until the final step.
- For the Cholesky factorization and its role in numerical linear algebra, see Strang [2016, Chapter 6]. The factorization is the computational backbone of Gaussian process inference in ML, where the kernel matrix is symmetric positive definite.
- Petersen and Pedersen [2012] provides a comprehensive collection of matrix identities, including derivative formulas for determinants, traces, and matrix inverses, which are essential for implementing gradient computations in neural networks.

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